

# Gemma3

## Gemma3

- 1. Model Scale
- 2. Performance
- 3. Using Gemma3
  - 3.1 Running Gemma3
  - 3.2 Having a Conversation
  - 3.3 Vision Capabilities
  - 3.4 Ending the Conversation
  - 3.5 Chinese Conversation
- References

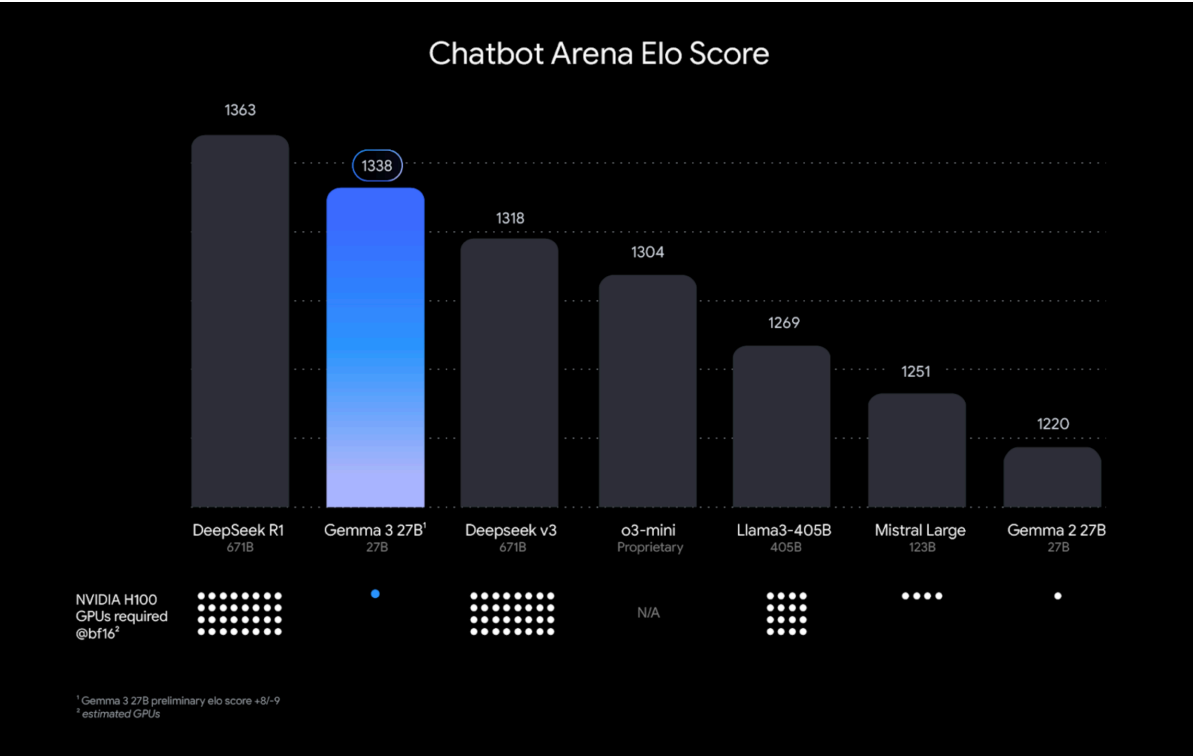
**Note: The RDK X5 4GB version needs to run a small parameter version due to performance limitations.**

Gemma is a family of lightweight models built by Google from the same research and technology used to create the Gemini models. The Gemma 3 model is a multimodal model (can process text and images) and has a 128K context window, supporting over 140 languages.

## 1. Model Scale

| Model     | Size  |
|-----------|-------|
| gemma3:1b | 815MB |
| gemma3:4b | 3.3GB |

## 2. Performance



## 3. Using Gemma3

### 3.1 Running Gemma3

Use the run command to start the model. If you have not downloaded this model before, it will be automatically pulled from the Ollama model library:

```
ollama run gemma3:4b
```

```
root@root:~$ ollama run gemma3:4b
pulling manifest
pulling aeda25e63ebd: 100% 3.3 GB
pulling e0a42594d802: 100% 358 B
pulling dd084c7d92a3: 100% 8.4 KB
pulling 3116c5225075: 100% 77 B
pulling b6ae5839783f: 100% 489 B
verifying sha256 digest
writing manifest
success
>>> Send a message (/? for help)
```

### 3.2 Having a Conversation

```
How to learn a programming language?
```

The response time depends on the hardware configuration, please be patient!

```
>>> How to learn a programming language?
Learning a programming language can feel daunting at first, but with a
structured approach and consistent effort, it's absolutely achievable.
Here's a breakdown of how to learn a programming language effectively,
broken down into stages and with different learning styles in mind:

**1. Choosing Your Language:**

* **Beginner-Friendly Options:**
  * **Python:** Highly recommended for beginners. It has a clean
    syntax, huge community support, and is used in many fields (web
    development, data science, scripting).
  * **JavaScript:** Essential for web development (front-end and
    increasingly back-end with Node.js).
  * **Scratch:** (Visual, block-based) A fantastic starting point for
    kids and absolute beginners to understand programming concepts without
    worrying about syntax.
* **Consider Your Goals:** What do you want to *do* with programming?
  * **Web Development:** JavaScript, Python (with frameworks like Django
    or Flask), Ruby
  * **Data Science/Machine Learning:** Python, R
  * **Game Development:** C#, C++, Python (with Pygame)
  * **Mobile App Development:** Swift (iOS), Kotlin/Java (Android)
```

### 3.3 Vision Capabilities



what do you see in this picture? ../test\_pic.png  
#In a conversation, use ": + the path to the image" to enable the model's vision capabilities and have it analyze the information in the picture.

```
root@localhost:~$ ollama run gemma3:4b
>>> What do you see in this picture?:../test_pic.png
Added image '../test_pic.png'
Here's a breakdown of what I see in the picture:

*   **Main Subject:** A young man is in the foreground, with his eyes closed and his head tilted back, seemingly taking a deep breath. He's wearing a white tank top and shorts.
*   **Background:** The background is a cityscape, likely a rooftop with buildings visible. There is a water bottle next to him.
*   **Other Person:** There's another man sitting on the right, also seemingly relaxed.
*   **Sky:** The sky is a vibrant blue with fluffy white clouds.
*   **Color Tone:** The image has a slightly warm, saturated color tone, giving it a dramatic feel.

This photograph evokes a sense of stillness, perhaps a moment of respite amidst a busy urban environment. It's a strong, visually compelling image.
>>>  Send a message (/? for help)
```

### 3.4 Ending the Conversation

Use the `Ctrl+d` shortcut or `/bye` to end the conversation!

### 3.5 Chinese Conversation

If you don't have a Chinese input method, you can refer to the tutorial on switching Chinese input methods.

Chinese conversation:

```
root@kali:~# ollama run gemma3:4b
>>> 一个小时是多少分钟？请用中文告诉我
一个小时有 60 分钟。
```

## References

---

### Ollama

Official Website: <https://ollama.com/>

GitHub: <https://github.com/ollama/ollama>

### Gemma3

Ollama Corresponding Model: <https://ollama.com/library/gemma3>